

THE RELATIONSHIP BETWEEN DEVICE ACCESSIBILITY AND CYBERLOAFING BEHAVIOR OF IC STUDENTS

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ABSTRACT

This study explored the relationship between device accessibility and cyberloafing behavior among IC students, recognizing the increase of reliance on digital devices for academic tasks and the growing concern about non-academic online activities among learners. The purpose of this research was to determine the students' level of device accessibility, assess their level of cyberloafing behavior, and examine whether a significant relationship exists between the two variables. Using a descriptive-correlational research design, the researchers gathered data through a structured survey that measured three dimensions of device accessibility—device ownership and availability, access to peripherals and tools, and internet and multimedia access—and three dimensions of cyberloafing behavior, including social media interaction, accessing online content, and gaming/gambling activities. Results revealed that the level of device accessibility among IC students was high, with an overall mean of 3.71, while the level of cyberloafing behavior was also high, with an overall mean of 3.88. The highest form of cyberloafing was accessing online content, indicating that students often view non-academic material during academic tasks. Correlational analysis showed a significant positive relationship between device accessibility and cyberloafing behavior ($r = 0.427$, $p = 0.001$), with device accessibility explaining 18.2% of the variance in cyberloafing. These findings suggest that greater access to digital devices and internet resources increases the ease and likelihood of engaging in cyberloafing, which aligns with the Theory of Planned Behavior wherein perceived behavioral control influences online actions. The results highlight the importance of promoting responsible digital behavior and implementing strategies that help reduce distractions in learning environments.

Chapter I

INTRODUCTION

Background of the Study

In today's digital age, it is common for students to bring internet-connected devices like smartphones, tablets, and laptops to school. However, students can use these devices for non-academic purposes, which can distract them from their lessons and frustrate instructors. This online engagement for non-academic purposes is called cyberloafing. Cyberloafing negatively affects the learning environment by reducing the focus and productivity of students during class [1].

Access to technological innovations like the Internet and mobile devices has brought both benefits and struggles in education. Once seen only in workplaces, cyberloafing or cyberslacking has now become common among students who are using devices meant for studying to multitask and engage in unrelated activities [2]. With the growing accessibility of Information and Communication Technology (ICT), learners become more prone to behaviors that decrease academic efficiency and promote digital distractions [3].

A study at Ateneo de Davao revealed that the students' reason for using their devices and social media is more for their social needs than academic purposes. Many of them spend time online to chat, play games, and watch videos rather than to accomplish their assignments. When accessing online materials, their attention is often diverted to social networking sites, which leads to decreased schoolwork focus [4]. Similarly, research from Davao del Norte State College found that access and exposure to smartphones and computers contributes to frequent social media use. The study emphasized the importance of comprehending how device use affects learning outcomes [5].

In addition, research in Capiz reported that students who are active on social media through cyberloafing usually overlook its negative effects. Although this behavior caused both positive and negative aspects, it may still hinder one's learning capacity and focus [6]. Another study on young Filipinos noted that while the Internet helps greatly in research and communication, excessive use can influence academic engagement and shape online identities that distract from schoolwork [7]. Moreover, research conducted at a university in Iloilo discovered that students' access to devices allows frequent use of social media for communication and information sharing. The findings indicated that students' attitudes toward social media influence their behavior, and that unethical online activities are linked to how they access and use their devices [8].

According to a study in Ankara, Turkey, it was determined that mobile phones have evolved from being used only as communication tools into platforms for entertainment and social interaction. This shift in usage patterns has contributed to problematic behaviors such as cyberloafing and decreased academic focus [9]. Likewise, a study in Nigeria emphasized that while most students access the Internet through their personal devices, they often use this access for leisure and entertainment rather than for academic tasks. The convenience of having access to device tends to draw them toward non-academic online activities, resulting in reduced concentration on school-related responsibilities [10].

While most studies have explored the benefits and disadvantages of device access in education, there is still limited research describing how students' cyberloafing behavior is influenced by device accessibility. Secondly, most existing studies have examined cyberloafing and device misuse of the general student population's cyberloafing behavior, but not specifically among students from the Institute of computing. Finally, there is a noticeable lack of research in the Philippines focusing on the relationship between Device Accessibility and

Cyberloafing Behavior. For these reasons, there is a clear need to investigate the relationship between these two variables.

Given these gaps, this study is significant because it aims to provide insights into how device accessibility influences the cyberloafing behavior of IC students. Particularly, the study aims to accomplish the following objectives:

General Objective:

To determine the relationship between device accessibility and cyberloafing behavior of IC students.

Specific Objectives:

1. To assess the level of device accessibility among the students of the Institute of Computing.
2. To determine the intensity of cyberloafing behavior displayed by IC students.
3. To examine the relationship between device accessibility and cyberloafing behavior.
4. To produce insights that may help administrators manage the use of devices and minimize students' cyberloafing behavior.

The results of this study can serve as a basis for formulating strategies to promote responsible device usage among students. Additionally, it can contribute to reducing distractions in academic settings and improving students' focus and performance.

Theoretical Framework

This present study is grounded in the Theory of planned behavior. According to the Theory of Planned Behavior, proposed by Ajzen (1985) and developed in 1987, human behavior is guided by three kinds of considerations: beliefs about the consequences of one's action, beliefs about what other thinks one should do, and about one's ability to carry out the action [11]. The beliefs that were mentioned shape a person's perception toward the behavior, the subjective

norm, and their behavioral control, all of which helps the formation of intention [12]. These beliefs provide the basis for the formation of a person's intention to engage in a behavior, but performing the behavior depends on the individual's behavioral control [11].

In the context of this study, a student's intention to be involved in cyberloafing can be stronger if they see it positively, believe that people around them consider it to be acceptable, and feel that they have the ability, opportunity and access to device. These intentions, combined with actual access to devices and the internet, increases the possibility of cyberloafing.

While the theory of planned behavior has been extensively applied in previous study involving health-related behaviors, academic misconduct, and adoption of technology, it has been rarely used to explore device accessibility and its relationship with students' cyberloafing.

Conceptual Framework

Figure 1 shows the conceptual model used in the present study. It explores the relationship between device accessibility and cyberloafing behavior among IC students. The purpose of the framework is to understand how access to mobile devices and the internet influences students' motivation to be involved in cyberloafing activities. The study adopts the Criterion-Predictor Model (CPM) since it is suitable for examining how one or more predictor variables influence a specific outcome or criterion. Previous research has shown that this model effectively determines the relationship between predictor and criterion variables, supporting its use in analyzing the connection between device accessibility and cyberloafing behavior in the present study [13].

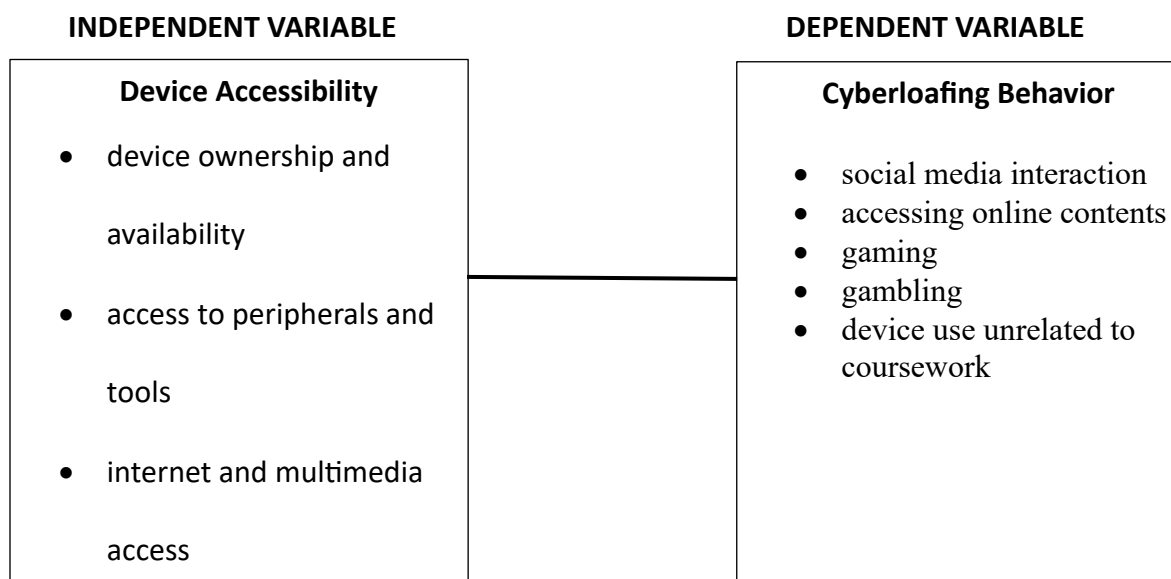


Figure 1: Conceptual Framework

Research Questions

This study intends to determine the relationship between device accessibility and cyberloafing behavior among IC students of Davao del Norte State College. Specifically, it seeks to answer the following:

1. What is the level of device accessibility among IC students in terms of:
 - 1.1 device ownership and availability;
 - 1.2 access to peripherals and tools;
 - 1.3 internet and multimedia access?
2. What is the level of cyberloafing behavior among IC students in terms of:
 - 2.1 social media interaction;
 - 2.2 accessing online contents;
 - 2.3 gaming/gambling purposes?

Null Hypothesis

Ho1: There is no relationship between device accessibility and cyberloafing behavior.

Chapter II

METHODOLOGY

Presented in this chapter are the following parts of the study: research design, research locale, participants of the study, sampling techniques, statistical treatments, data collection procedure, research instruments, and ethical considerations.

Research Design

The study adopts a descriptive-correlation research design, which is appropriate for examining the relationship between device accessibility and cyberloafing between device accessibility and cyberloafing behavior among IC students, supported by the study of Aciksoz et al. [14], who employed a descriptive-correlational and cross-sectional approach to explore the relationship between cyberloafing and innovativeness among nurses in research hospital. The reason for its appropriateness is its ability to enable description of existing conditions and determining the degree of association between variables without manipulating them, which makes it ideal for the investigation of naturally occurring behaviors like technology use [15]. Additionally, the descriptive-correlational approach has been successfully employed by previous research in exploring the association between cyberloafing, digital involvement, and psychological or academic factors, thus, validating its effectiveness for similar studies [16]. Furthermore, a local study used a correlation approach to determine the relationship between demographic factors on online learning engagement, supporting the use of this design in academic context [17]. Likewise, several international studies implemented the same design to examine the link between cyberloafing and social media addiction, reinforcing its appropriateness to the present research [18].

Research Locale

This research will be conducted at Davao del Norte State College (DNSC). DNSC is an institute of free higher education that started as Davao del Norte School of Fisheries, located in Barangay New Visayas, Panabo City, Davao del Norte, Philippines. It offers various programs that are divided into four Institutes, and is well known for its commitment to technological and academic excellence. Specifically, the study focuses on students from the Institute of Computing, which offers the programs Information Technology and Information Systems.

Participants of the Study

Participants in this study are composed of students from the Institute of Computing (IC) at Davao del Norte State College (DNSC). Specifically, the students from IC were chosen as the respondents of this study because of their high exposure to digital devices and the internet, which they frequently use for their online-based academic tasks. Therefore, they are considered highly relevant for the study on device accessibility and cyberloafing behavior. Overall, the participants represent a diverse group with varying levels of device accessibility and cyberloafing behavior, which is essential for achieving the aims of the study.

Sampling Techniques

The target population of this study was all students from the Institute of Computing enrolled at DNSC. To make sure the sample represented the population of both program from the Institute of Computing, stratified random sampling was used. Stratified random sampling is a probability sampling technique that involves dividing the target population into homogeneous and mutually exclusive subgroups, known as strata. From each stratum, a simple random sample is selected, then the samples are combined as the overall sample [19]. In this study, the two strata were the Bachelor of Information Technology (IT) and Bachelor of Information

Systems (IS) students under the Institute of Computing, and 158 participants were randomly selected from each until the desired number of participants was reached.

In total, 316 students will take part in the study, including both male and female students as well as students from all year levels. The sample size was determined using Slovin's formula, computed using the total number of IC students, from the DNSC Registrar report on Enrollment summary for first semester 2025-2026. This approach ensured balance representation between the two programs.

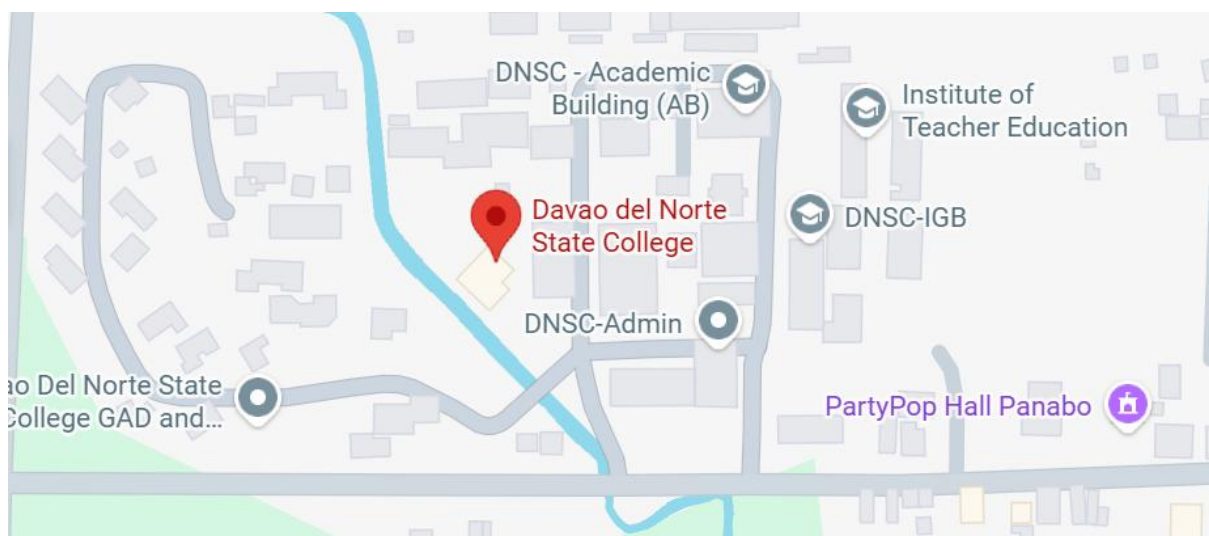


Figure 2: Location of Davao del Norte State College

Statistical Treatments

For more comprehensive interpretation and analysis of the data, this study utilized frequency and percentage, mean and standard deviation, and Pearson's r as the main statistical tools. Since both independent and dependent variable were measure with 5-point Likert scales, these statistical treatments are appropriate and mostly used in descriptive-correlation studies [20]. Firstly, frequency and percentage presented the respondents' demographic profile and the distribution of responses for each question [21]. Secondly, mean and standard deviation summarized the variability of device accessibility and cyberloafing behavior. Whereas, Likert-type data can be treated as approximately interval in nature for such reasons [22].

Finally, Pearson's r measures the strength and direction of linear association between two continuous variables. For this reason, it was used to determine the relationship between the two variables [23]. Previous studies on cyberloafing and technology-related behaviors have also applied the previously mentioned statistical techniques, confirming their suitability and reliability [24]. Therefore, these methods were deemed appropriate in producing valid, interpretable and evidence-based findings regarding the relationship between device accessibility and cyberloafing behavior among IC students.

Data Collection Procedure

The researcher first prepared a formal written communication addressed to the validators for the review and evaluation of the research questionnaires. These instruments were subjected to feedback, including comments, suggestions, and recommendations. Subsequently, a letter will be sent to the Dean of the Institute of Computing. The purpose of the letter is to request permission to conduct the study. Once the approval of the dean is acquired, the researchers will ensure that the rules and conditions will be followed. The questionnaires will then be distributed to the respondents, the Institute of Computing Students. The collected data was compiled, tabulated, analyzed, and interpreted in confidence, following the guidance of a statistician based on the study's objectives.

Finally, after the collected data served its research purpose, all collected data were deleted from the database after analysis. This is to ensure further protection of respondents and to uphold their safety and privacy.

Research Instruments

In collecting data for this study, survey questionnaires were distributed to the respondents as the primary instrument. Two sets of questionnaires were utilized to obtain relevant data to

the independent and dependent variables. One questionnaire is for device accessibility, while the other one is for cyberloafing behavior.

Device Accessibility. The questionnaire for Device Accessibility was adapted from the study Device Ownership, Accessibility, and University Students' Perceptions of an Online Foundation English Course in Thailand [25]. The questionnaire includes a total of 15 items, and is consisted of three parts, namely: device ownership and availability (1-5), peripherals and tools (6-10) and peripherals and tools (11-15).

Cyberloafing Behavior. The second questionnaire was used to determine the level of cyberloafing behavior of IC students. It is adapted from the study of Koay Kian Yeik on Assessing Cyberloafing Behavior among University Students [26]. Additionally, the questionnaire has 15 items in total, with 3 sections: social media interaction (items 1-5), accessing online content (items 6-10), and gaming or gambling purpose cyberloafing (items 11-15).

All of the instruments used in this study were contextualized to ensure that they align with the objectives of the study. The feedback and recommendations from the adviser of the course, members of the panel, and professional validators were applied into the final version of the tools to ensure clarity, relevance, and construct validity.

Ethical Considerations

In qualitative research, there are three ethical principles to ensure the protection and fair treatment of participants, namely **autonomy, beneficence, and justice**. Autonomy refers to the recognition of the participants' rights to make informed and voluntary decisions, including the responsibility of the researchers to provide clear information about the study, obtain informed consent, and allowing participants to withdraw at any time without consequence.

Meanwhile, beneficence is centered on promoting the well-being of the respondents by doing good and avoiding harm, researchers must take steps to protect their privacy and confidentiality and consider how findings are presented to prevent unintended identification. Finally, justice emphasizes fairness and equality by ensuring that all participants are treated with respect, calling for sensitivity to their circumstances and recognizing their value. Altogether, these three guide researchers to conduct a study that is ethical, upholding integrity, accountability, and respect for human dignity [27].

Chapter III

RESULTS AND DISCUSSIONS

Results

RQ1. Level of Device Accessibility Among IC Students

Table 1. Device Accessibility in Terms of Device Ownership and Availability

As shown in Table 1, IC students demonstrated a generally positive level of device ownership and availability ($M = 3.52$, Agree). This means that most students have at least one device accessible for academic use, although some limitations exist such as outdated units or lack of multiple devices.

Table 2. Device Accessibility in Terms of Access to Peripherals and Tools

Table 2 shows that students have sufficient access to peripherals such as keyboards, chargers, headphones, and storage devices ($M = 3.79$, Agree).

Table 3. Device Accessibility in Terms of Internet and Multimedia Access

Table 3 shows a generally positive level of internet and multimedia access ($M = 3.83$, Agree). Students strongly agreed that they have reliable connections for academic and communication needs.

Table 4. Summary of Device Accessibility

Overall, Table 4 indicates that IC students have a high level of device accessibility ($M = 3.71$, Agree). This suggests that most students can adequately participate in online tasks and digital learning. Findings reflect trends in modern educational settings where access to technology is increasingly normalized, but still varies in quality across individuals.

RQ2. Level of Cyberloafing Behavior Among IC Students

Table 5. Cyberloafing Behavior: Social Media Interaction

Table 5 shows that IC students frequently engage in social media interaction during academic tasks ($M = 3.92$, Agree). TPB suggests that this behavior becomes habitual when students believe such actions are easy and readily available.

Table 6. Cyberloafing Behavior: Accessing Online Content

Students scored highest in accessing non-academic online content ($M = 4.24$, Strongly Agree). The ease of access increases behavioral intention, which TPB identifies as a major predictor of engagement in off-task online activities.

Table 7. Cyberloafing Behavior: Gaming/Gambling Purposes

Table 7 reveals moderate levels of gaming-related cyberloafing ($M = 3.50$, Agree), while gambling behaviors scored low. This finding shows that gaming is highly accessible and reinforcing, which strengthens behavioral intention per TPB.

Table 8. Summary of Cyberloafing Behavior

Overall cyberloafing behavior among IC students is high ($M = 3.88$, Agree). Consistent with TPB, this behavior is shaped by device accessibility, behavioral intention, and perceived ease of performing the behavior.

RQ3. Relationship Between Device Accessibility and Cyberloafing Behavior

Table 9. Correlation Between Device Accessibility and Cyberloafing

Table 9 shows a significant positive relationship between device accessibility and cyberloafing behavior ($r = 0.427$, $p = 0.001$). This indicates that higher accessibility increases the likelihood of cyberloafing. The coefficient of determination ($r^2 = 0.182$) shows that device accessibility explains 18.2% of the variance in cyberloafing.

Discussions

The results of this study indicate that IC students have a generally high level of device accessibility, encompassing device ownership, access to peripherals, and reliable internet connectivity. This aligns with current educational trends where students increasingly rely on digital tools for learning, communication, and entertainment. The high accessibility provides students with the necessary resources to engage in online activities efficiently, which, as the results show, includes both academic and non-academic behaviors.

The study also revealed that cyberloafing behavior among IC students is notably high, particularly in accessing social media and non-academic online content. This finding is consistent with previous research indicating that students with greater internet skills and device access are more likely to engage in off-task online behaviors. For example, studies reported that students with higher Internet competence tend to cyberloaf more [28], and that internet skills are significantly associated with socialisation-related cyberloafing [29]. These studies support the idea that device accessibility, combined with competence, can facilitate more frequent engagement in non-academic online activities.

The positive correlation between device accessibility and cyberloafing behavior ($r = 0.427$, $p = 0.001$) further confirms this relationship. As accessibility improves, students have

easier and faster means to engage in cyberloafing, which reflects the theoretical framework of the Theory of Planned Behavior (TPB), where ease of access and perceived behavioral control influence intention and actual behavior. This relationship is also echoed in the findings where it observed that students with higher self-efficacy in internet use are more likely to engage in cyberloafing [30], suggesting that access alone is a facilitator but personal confidence and attitudes toward the behavior play a critical role as well.

Moreover, cognitive absorption appears to mediate this relationship. Previous studies show that deep engagement with digital devices can enhance the likelihood of cyberloafing [31] [32]. When students are absorbed in their devices, especially those with high accessibility, they are more susceptible to distractions from social media, entertainment, and other non-academic content. Similarly, it was highlighted that frequent internet use fosters habitual cyberloafing, reinforcing the notion that accessibility facilitates engagement in off-task behaviors [33].

Overall, the findings suggest that while high device accessibility enables effective participation in digital learning, it also increases the risk of cyberloafing. The results underscore the importance of designing interventions that balance accessibility with responsible digital behavior, such as promoting self-regulation, raising awareness of online distractions, and integrating engaging academic digital activities that can capture students' attention constructively.

Chapter IV

CONCLUSIONS AND RECOMENDATIONS

Conclusions

The study concludes that the students from the institute of computing have a high level of device accessibility, enabling them to participate effectively in digital academic tasks. However, this accessibility also comes with high levels of cyberloafing, especially in browsing online content and interacting on social media. A significant positive relationship was found between device accessibility and cyberloafing behavior, indicating that convenient access to devices increases the likelihood of off-task digital activities. These findings align with the Theory of Planned Behavior, suggesting that when students perceive cyberloafing as convenient and easy, their intention to engage in it becomes stronger

Recommendations

Based on the results, it is recommended that students practice responsible device use and develop self-regulation strategies to limit cyberloafing during academic tasks. Instructors are encouraged to implement engaging activities and clear guidelines to reduce opportunities for off-task behavior. School administrators should promote digital discipline programs that teach responsible online behavior while supporting students with necessary technological resources. Future research may explore other factors that influence cyberloafing, such as motivation, environment, or self-control.

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